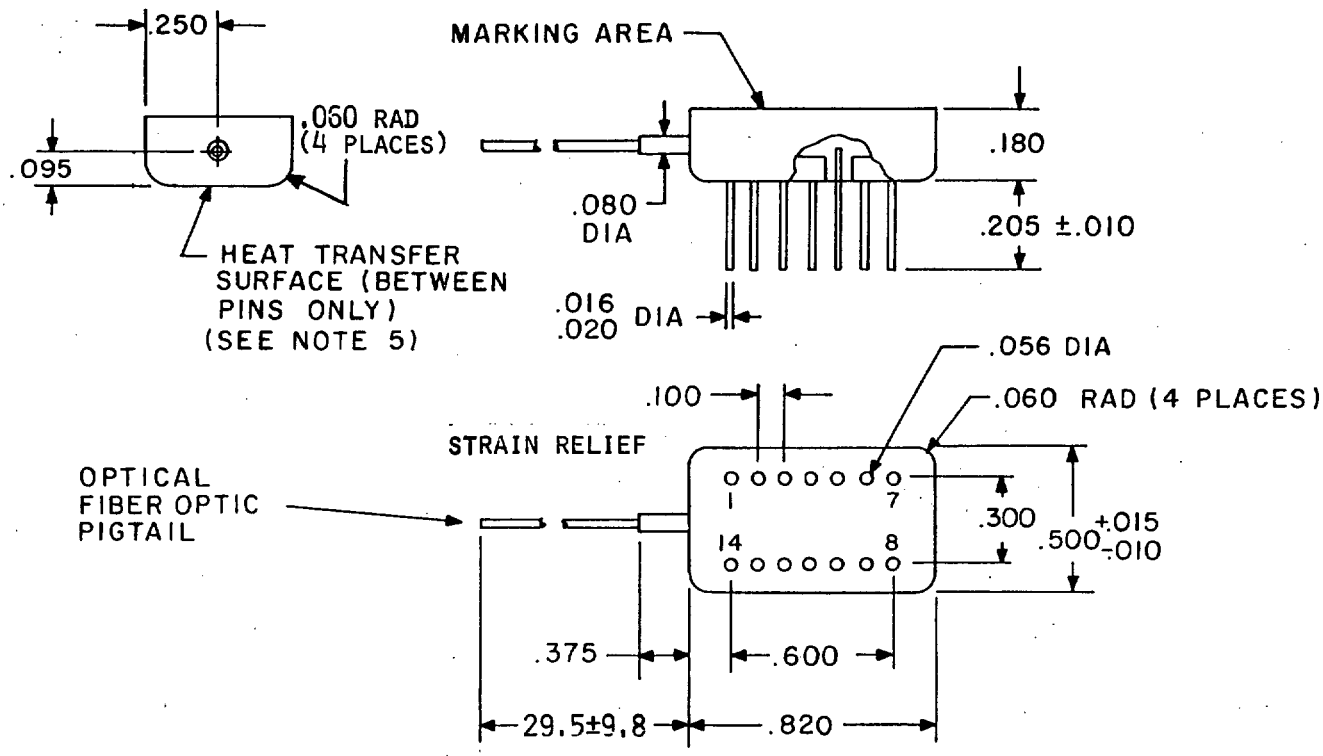
MILITARY SPECIFICATION SHEET

DETECTOR, PINFET, FIBER OPTIC, 1100-1600 NANOMETERS, GLASS PIGTAILED, HERMETICALLY SEALED, DUAL IN-LINE PACKAGE (DIP)

This specification is approved for use within the Communications and Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the fiber optic detector described herein shall consist of this specification and the latest issue of DOD-D-24620.



Inches	mm	Inches	mm
.010	0.25	.205	5.21
.015	0.38	.250	6.35
.016	0.41	.300	7.62
.020	0.51	.375	9.53
.056	1.42	.500	12.70
.060	1.52	.600	15.24
.080	2.03	.820	20.83
.095	2.41	9.8	248.92
.100	2.54	29.5	749.30
.180	4.57		

FIGURE 1. PINFET detector - (DIP) package outline and pin connections.

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Pin connections

Pin number	Function
1	Detector bias (-5 V)
2	NC
3	Ground
4	-5 V supply ±0.5 V
5	Ground
6	NC
7	Output (must be ac coupled)
8	Ground
9	NC
10	+5 V supply ±0.5 V
11	NC
12	NC
13	AGC control
14	NC

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm).
4. Leads shall be gold (50 microinches min.) per MIL-G-45204, type II, grade A, class 1 over nickel (100 microinches min.) per QQ-N-290, class 1, grade C or MIL-C-26074.
5. The heat transfer surface (heatsink) shall provide the major heat transfer mechanism to remove heat from the device. Convection and radiation to the environment surrounding the device must not provide significant heat transfer for this device when it is being tested to meet this specification.
6. A yellow band with the phrase "fiber optic" included within it shall be provided on the strain relief of the fiber optic pigtail.

FIGURE 1. PINFET detector - (DIP) package outline and pin connections.- Continued.

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REQUIREMENTS:

Design and construction: See figure 1.

Pigtail fibers: As specified by DOD-F-49291/1 with the exception that the buffer coating be upjacketed to a diameter of 975 µm ±50 µm. The fiber end termination shall be unconnectorized.

Fiber retention:

Tension: Fiber pigtails shall withstand a minimum tensile load of 5 newtons for a minimum of 30 seconds.

Environmental requirements:

Temperature range:

Operating: -46°C to +85°C.

Storage: -57°C to +85°C.

Barometric pressure:

Operating: Sea level up to 10,000 feet (3,050 m).

Storage and transit: Sea level up to 40,000 feet (12,200 m).

Nuclear radiation resistance: The PINFET device shall be designed to withstand the nuclear radiation requirements for RHA designation M of MIL-S-19500. When measured 15 minutes after the specified irradiation, the detector shall satisfy all the requirements contained herein except as specified in table I.

Electrical and optical requirements: See table II.

Marking: Marking shall be in accordance with DOD-D-24620 with the following additions:

a. Lot identification code.

b. Serial number.

c. Date code.

Screening: Screening shall be conducted on 100 percent of all devices prior to groups A, B, and C inspections and shall be in accordance with the methods specified in table III. Tests shall be performed in accordance with method 5008 of MIL-STD-883.

Group A inspection: Group A inspection shall consist of the examinations and tests specified in table IV.

Group B inspection: Group B inspection shall consist of the examinations and tests specified in table V.

Group C inspection: Group C inspection shall consist of the examinations and tests specified in table VI.

Part number: M24620/4-001.

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TABLE I. Post nuclear requirements. 1/

Responsivity R (V/μs)	Output noise voltage (V_N)	
	Temperature range	Noise limit
Minimum 94 percent of table II	-46°C to +22°C	285 μV
	23°C to +35°C	317 μV
	36°C to +80°C	420 μV

1/ The detector module shall satisfy all the requirements previously specified except for those parameters stated in this table after nuclear exposure.

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TABLE II. Electrical and optical requirements.

Detector dash number M24620/4-	Optical	Electro-optical						Electrical			
	Spectral sensitivity λp (nm)	AGC control voltage 1/	Input optical power (dBm)	Modular responsivity R (V/W) 2/		Dynamic range (dB) for 10^{-9} BER	Responsivity flatness R_F (dB)	Load resistance (ohms)	Output noise 3/ voltage $V_N(\mu V)$		Total power 4/ consumption (mW)
				DC to 3 MHz	3 MHz to 20 MHz				Temperature range	Noise limit	
001	1200 to 1600	AGC off	-44.5 to -7.5	$9(10^4)$ to $12(10^4)$	$5(10^4)$ to $12(10^4)$	37	±1.0	≥ 1000 AC coupled	-46°C to 22°C	190	< 175 for each device
									22°C to 35°C	200	
		AGC on		$5.3(10^3)$ to $12(10^3)$	$3.3(10^3)$ to $12(10^3)$				36°C to 85°C	265	< 50 from negative supply

1/ The AGC control voltage shall be -4.75 V dc for full off and -2.00 V dc for full on.
2/ The module responsivity shall be valid over the input optical ranges specified for the AGC conditions.
3/ V_N is the total RMS noise voltage appearing across the load resistor for a measurement bandwidth of 100 kHz to 11 MHz. The low pass characteristics of the noise filter shall be that of a 2 pole Butterworth having a -3 db corner at no less than 11 MHz.
4/ The normal operating voltages shall be applied to the module and the input currents shall be measured. This test shall be performed with the fiber optic pigtail input covered.

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TABLE III. Screening inspection.

Screen inspection (100%)	MIL-STD-883 method	Details
Internal visual inspection	2017	Ambient
Stabilization bake	1008	24 hours minimum at T = 85°C
Thermal shock (temperature cycling)	1010	Test condition A, steps 1 and 3 shall be 2 hours minimum
Constant acceleration	2001	Y1 orientation only at 5,000 G minimum
Hermetic seal Fine leak rate Gross leak rate	1014	Test condition A_1, 5×10^{-7} atm cm^3/sec maximum leak rate. Test condition C, except that leak indicator fluid shall be maintained at 100°C ±5°C
Shock	2002	Half-sine pulse at 100 G's
Steady-state burn-in	1015	Condition C for 168 hours minimum, temperature (heat sink) = 85°C
External visual and mechanical inspection	2009	Ambient

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TABLE IV. Group A inspection. 1/

Inspection	Details	LTPD
Subgroup 1 2/		
Module responsivity	25°C	5
Frequency response		5
Output noise		5
Dynamic range		5
Power consumption		5
Subgroup 2		
Module responsivity	-46°C	10
Output noise		10
Power consumption		10
Subgroup 3		
Module responsitivity	85°C	10
Output noise		10
Power consumption		10
Subgroup 4		
Spectral Sensitivity	25°C	15
Power supply 2/		15

1/ All measurements except for spectral sensitivity shall be made at a wavelength of 1290nm ± 20nm.

2/ The modules shall meet the responsivity and output noise requirements of subgroup 1 for the following conditions of power supply voltages:

Condition	Positive supply	Negative supply
A	+5.50	-5.50
B	+5.50	-4.50
C	+4.50	-5.50
D	+4.50	-4.50

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TABLE V. Group B inspection.

Inspection	Method	MIL-STD-883 details	LTPD
Subgroup 1 1/			
Solderability 2/	2003		15
Lead integrity	2004	Test condition C_1	15
Resistance to solvents	2015		15
Subgroup 2 1/			
Internal visual and mechanical	2014		15
Bond strength	2011		10
Die shear strength	2019		15
Subgroup 3			
Physical dimensions	2016		20
Fiber retention			10
Seal:	1014	Test condition A_1 for	10
Fine		fine leaks (leak testing	
Gross		2 hours after pressurization is acceptable);	
End points:		test condition C_1 for	
Subgroup 1		gross leaks, except that	
Group A		leak indicator fluid shall be maintained at 100°C ±5°C	

1/ Failed devices from the same inspection lot may be used for all subgroups when electrical or optical end-point measurements are not required.

2/ The LTPD for solderability tests applies to the number of leads inspected except in no case shall less than three devices be used to provide the number of leads required.

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TABLE VI. Group C inspection.

Inspection	Method	MIL-STD-883 details	LTPD
Subgroup 1			
Thermal shock	1010	Test condition A_1, except the minimum time for steps 1 and 3 shall be 2 hours.	10
Moisture resistance	1004		10
Hermetic seal	1014	Test condition A_1 for fine leaks (leak testing 2 hours after pressurization is acceptable); test condition C for gross leaks, except that leak indicator fluid shall be maintained at 100°C ±5°C	10
External visual End points: Subgroup 1 Group A	2009		10
Subgroup 2			
Shock	2002	Non-operating; 350 G; t = 6 ms; 5 blows in each orientation	10
Constant acceleration End points: Subgroup 1 Group A	2001	Non-operating; 5000 G; in each orientation	10
Subgroup 3			
Barometric pressure End points: Subgroup 1 Group A	1001		15
Subgroup 4 1/			
Salt atmosphere	1009		15
Subgroup 5			
Steady state operation life End points: Subgroup 1 Group A	1005	T = 85°C Test condition B	7
Subgroup 6			
Nuclear survivability	1019 1017		
End points: Subgroup 1 Group A			15

1/ Failed devices from the same inspection lot may be used for all subgroups when electrical or optical end-point measurements are not required.

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Custodian:
Army - CR

Review activity:
DLA - ES

Preparing activity:
Army - CR

Agent:
DLA - ES

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